

Layer 5

Request life cycle

When Request hits LLM, it goes through a strict sequence of operation:

Request arrive $\rightarrow$ input tokenized $\rightarrow$ prompt or chat template is built

$\rightarrow$ Prefill Phase occurs $\rightarrow$ First token generated $\rightarrow$ Decode Loop begin

$\rightarrow$ output stream back to the user $\rightarrow$ Stop condition is met

Logged

Prefill vs Decode

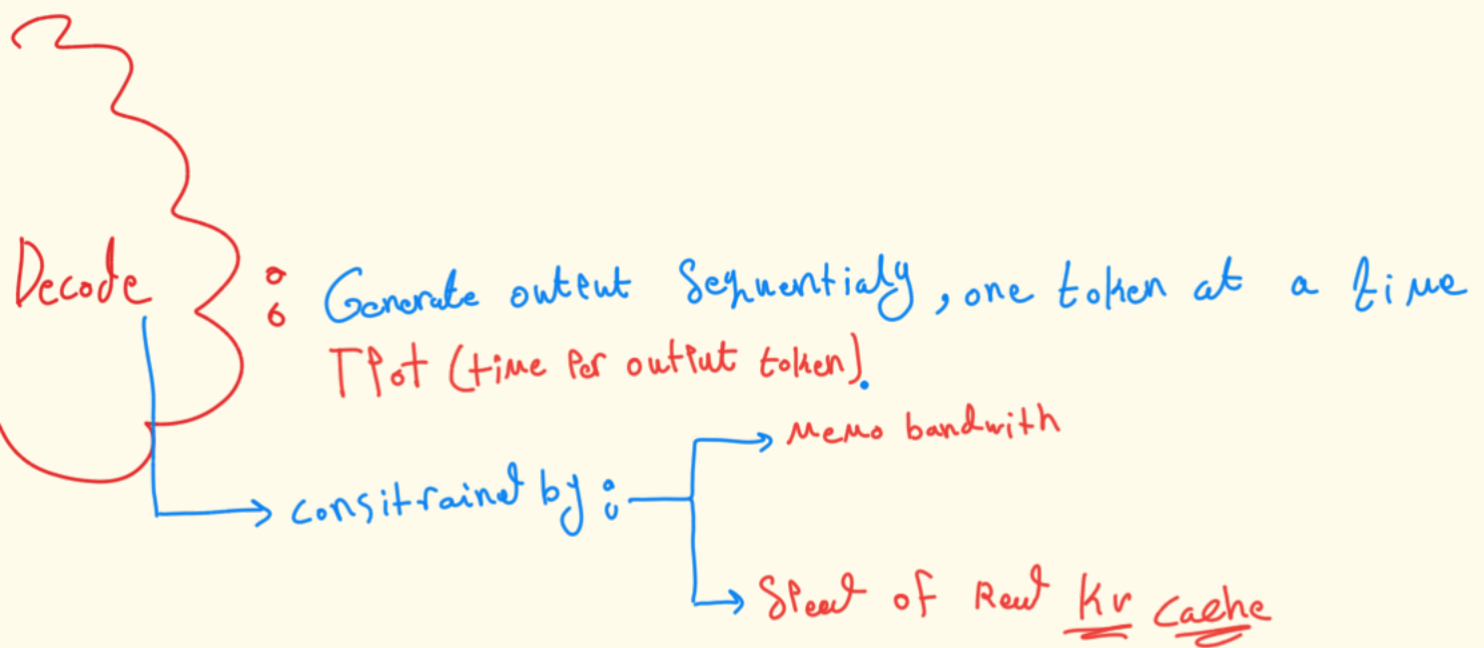
Prefill

Processing input prompt to create initial

KV's $\rightarrow$ TTFT (time to first token) a must to watch

$\uparrow$ Prompts

$\uparrow$ TTFT



Latency vs Throughput

how long user
will actually wait

a system may achieve
high throughput but
poor user latency

measures token
per sec

Batching
strategies

- efficiently utilize GPU's

— static batching: LLM Requests have highly variable lengths → wasting hardware capacity

— continuous batching: dynamic approach that → add
→ remove

Requests on the Fly

↳ improve utilization when handling live traffic

essential
metrics
to
track

- TTFT, TPoT, Latency, throughput
- (P50, P95, P99) Latency Percentiles,
- GPU and RAM utilization and usage
- Queue time, error rate

END
OF
inference
Fundamentals